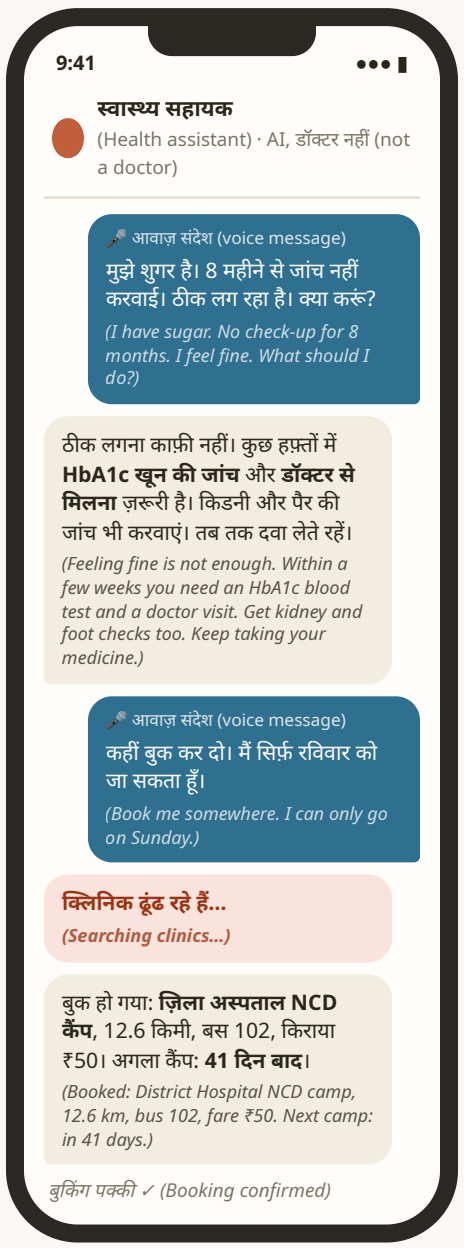
From a good answer to a reachable test

A 40-year-old factory worker with limited literacy, in a village in Faridabad district, asks an AI health app about his diabetes by voice, in Hindi. The answer is right. Whether he gets care depends on where the app routes him. Fictional case; facilities and prices are illustrative.

1 · THE ANSWER



2 · THE FRONT DOOR TO CARE

The app now chooses where he goes.

AI as router

It searches a directory, ranks options, shows a few, and books one.

Home	village, 20-km limit
Coverage	PM-JAY covers hospital stays, not this visit; he pays
Budget	₹600 for fee, test and fare
Work	daily-wage shifts, Mon–Sat
Transport	no vehicle, takes the bus
Deadline	seen within 30 days

DESTINATION
HbA1c test and doctor review for diabetes follow-up

3 · WHERE THE ROUTE CAN BREAK

c1 Unreachable from where he lives — ✕ Too far
The top-ranked hospital is 31 km away in the city. He can travel 20.

c2 Outside his coverage and budget — ✕ Over budget
Home sample collection plus teleconsult costs ₹1,148. PM-JAY does not cover outpatient tests.

c3 No HbA1c test nearby — ✕ No HbA1c
The village PHC, 2 km away, checks sugar with a glucometer but cannot run an HbA1c test.

c4 Can't get time off work — ✕ No Sunday slot
The block CHC's NCD clinic is free, but open Mon–Sat, 9 am to 2 pm, during his shifts.

c5 No bus home after the clinic — ✕ No bus
An evening diabetes clinic 17.8 km away runs 8 to 10 pm. The last bus back leaves at 7:30.

c6 Booked, but too late — ✕ Past deadline
Free, on a bus route, on a Sunday, but the next NCD camp is 41 days out. The app books it anyway.

c7 A route that works — ✓ In time
An urban PHC Sunday OPD, 14.2 km on a bus route, free test, slot in 6 days. It sits on page 2 of the results, below sponsored listings.

4 · THE AUDIT, WORKED OUT FOR HIM

FEASIBLE SET: THE CLINICS THAT WOULD ACTUALLY WORK

$$F_i = \{\, c \in S_i : d(c, h_i) \le m_i,\ w_c \le \tau,\ \pi_i(c) \le b_i,\ a_i(c) = 1 \,\}$$

S_i	clinics offering what he needs: an HbA1c test and a doctor review
$d(c, h_i) \le m_i$	distance from his home within his travel limit, 20 km
$w_c \le \tau$	next appointment within the clinical window, 30 days for overdue follow-up
$\pi_i(c) \le b_i$	his total cost (consultation + HbA1c + bus fare) within his budget; free at public facilities
$a_i(c) = 1$	meets his stated needs: on a bus route both ways, and open on Sunday

For him: m = 20 km, τ = 30 days, b = ₹600, a = bus route both ways ∧ Sunday hours. For c8: π = ₹150 + ₹380 + ₹50 = ₹580 ≤ ₹600.

WHAT THE APP DID, SCORED AGAINST F

$$R = |C \cap F| / |F| \quad D = |L \cap F| / |F| \quad E = 1[\text{booked} \in F]$$

C = clinics the search retrieved, L = clinics shown to him, E = the booking works

CLINIC	HBA1C + DOCTOR	KM ≤ 20	DAYS TO APPT ≤ 30	HIS COST ≤ ₹600	BUS ROUTE	HOURS SUNDAY?	IN F?	SEARCH RANK	SHOWN (L)
c1 City Multispeciality Hospital sponsored	yes	31.0	3	₹1,850	yes	Sun	no	1	✓
c2 HomeLab + teleconsult sponsored	yes	home	2	₹1,148	—	any	no	2	✓
c3 Village PHC	glucometer	2.1	·	·	·	·	no	not listed	
c4 Block CHC, NCD clinic	yes	9.4	5	₹40	yes	Mon–Sat	no	4	
c5 Evening Diabetes Clinic	yes	17.8	7	₹560	no	eve	no	6 · p.2	
c6 District Hospital NCD camp BOOKED	yes	12.6	41	₹50	yes	Sun	no	3	✓
c7 Urban PHC, Sunday OPD	yes	14.2	6	₹50	yes	Sun	yes	7 · p.2	
c8 Family clinic + partner lab	yes	11.3	9	₹580	yes	Sun	yes	5	

F = {c7, c8}, C = page 1 = {c1, c2, c6, c4, c8}
Page 1 holds 5 results, sponsored first.

R = |{c8}| / 2 = 0.5
D = |∅| / 2 = 0
It found one route that works, then showed neither.

E = 1[c6 ∈ F] = 0
Answer graded correct, route failed: he waits 41 days.

WHAT THE AUDIT ASKS
Grading the answer alone sees none of step 3.

Found it? Did the search retrieve the routes that work for him? RETRIEVAL, R

Showed it? Did he see a usable option, or only sponsored and unreachable ones? DISPLAY, D

Booked it? Does the booking meet his deadline, budget, travel and work limits? EXECUTION